warming, topsoil erosion, ozone layer depletion, ocean currents, acid rain, the effects of floods and droughts, and new dangers we haven't yet discovered. This is straightforward planetary hygiene.

Global positioning systems are now in place so that your locale is radio-triangulated by several satellites. Holding a small instrument the size of a modern shortwave radio, you can read out to high precision your latitude and longitude. No crashed airplane, no ship in fog and shoals, no driver in an unfamiliar city need ever be lost again.

Astronomical satellites peering outward from Earth's orbit observe with unsurpassed clarity—studying questions ranging from the possible existence of planets around nearby stars to the origin and fate of the Universe. Planetary probes from close range explore the gorgeous array of other worlds in our solar system, comparing their fates with ours.

All of these activities are forward-looking, hopeful, stirring, and cost-effective. None of them requires "manned"* spaceflight. A key issue facing the future of NASA and addressed in this book is whether the purported justifications for human spaceflight are coherent and sustainable. Is it worth the cost?

But first, let's consider the visions of a hopeful future vouchsafed by robot spacecraft out among the planets.

VOYAGER 1 AND *VOYAGER 2* are the ships that opened the Solar System for the human species, trailblazing a path for future generations. Before their launch, in August and September 1977, we were almost wholly ignorant about most of the planetary part of the Solar System. In the next dozen years, they provided our first detailed, close-up information on many new worlds—some of them previously known only as fuzzy disks in the eyepieces of ground-based telescopes, some merely as points of light, and some whose very existence was unsuspected. They are still returning reams of data.

These spacecraft have taught us about the wonders of other worlds, about the uniqueness and fragility of our own, about beginnings and ends. They have given us access to most of the Solar